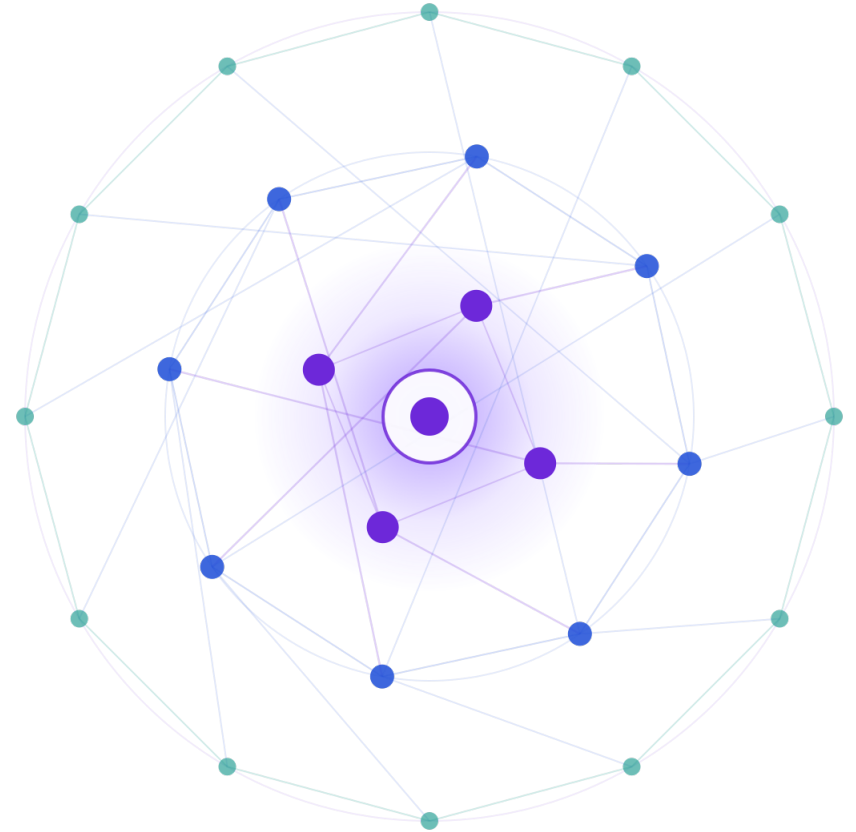


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 04 (Streaming Live RAG)
- Team Name - Team Sudarshan
- College Name - SRM Institute of Science and Technology, Kattankulathur
- Member Name & Email 1- Rachit Gandhi, rg7062@srmist.edu.in
- Member Name & Email 2- Vaibhav Naiyar, vn9026@srmist.edu.in
- Submission Github link - github.com/VaibhavNaiyar/Rag-bd | github.com/VaibhavNaiyar/Rag-fd



Theme

Theme 04: Streaming Live RAG

In a live voice conversation, batch RAG waits for the speaker to finish, embeds the whole utterance as one query, and starts over whenever a detail is added. The theme asks for an engine that retrieves while the person is still speaking, and measures every behaviour with its own telemetry.

01

Latency is structural

Waiting for end of speech adds the whole sentence length to every answer.

02

One utterance is not one query

“Venue, cancellation policy and catering” is three searches, not one blurry average.

03

A conversation edits itself

“Actually, the trip was international” should refine the existing answer, not restart it.

04

Never state what isn't verified

No fabricated citations, and no searching on “say that in two bullets”.

The bar: six acceptance gates

G1 Reproducibility

one command, no manual step

G2 Early retrieval

≥ 80% of eligible turns

G3 Multi-intent

≥ 70% isolate ≥ 2 intents

G4 Grounding

≥ 85% support, 0 fabricated

G5 Refinement

every late-detail turn

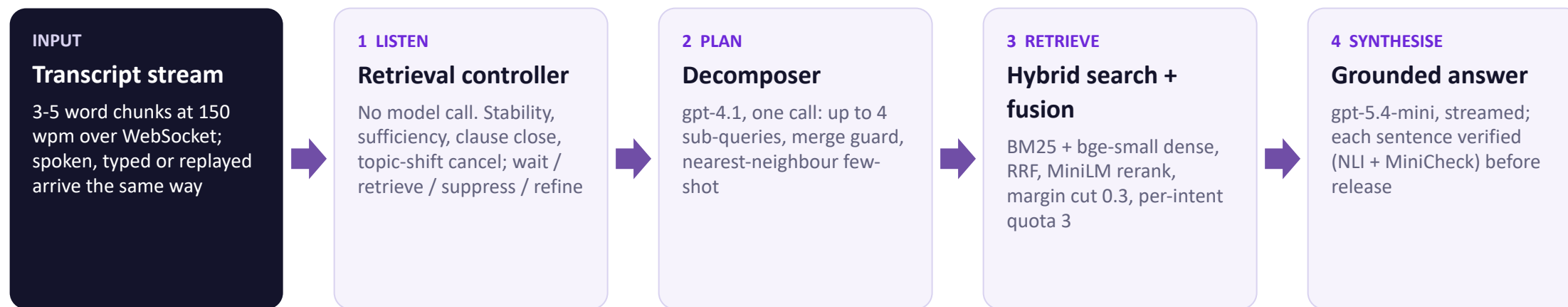
G6 Telemetry

100% complete traces

Existing Solutions & Gaps

Existing approach	Gap on live speech	What Streaming Live RAG does	Measured vs batch baseline
Batch “wait, then retrieve” RAG	Nothing happens while the user speaks	Rule controller fires cancellable provisional searches mid-utterance	Early retrieval 88.9% vs 0%
One query per utterance	The quietest need is starved in a blended search	Decomposer: up to 4 sub-queries, per-intent quota in fusion	Recall 85.5% vs 65.3% (ASQA)
Restart on every correction	Discards correct work; disjoint reply	Claim-level KEEP / EDIT / ADD / DROP; delta search only	Refined in place 100% vs 0%
Every turn triggers retrieval	“Two bullets, please” still hits the index	Suppression: presentation phrase + anaphora + no new content	Searches on reformat turns 0% vs 100%
Prompt-only “cite your sources”	Model-authored citations can be invented	Markers built server-side; each sentence verified before it streams	0 fabricated shipped in 142 turns
Agent frameworks, LLM per chunk	Model calls and latency on the hot path	Plain asyncio, ~39 ms rule decision per chunk, 2 LLM calls by design	LLM controller: 0% early vs 79.5%

Our Solutions & Architecture Diagram



Output validated sentences with server-built citations | uncertainty instead of guesses | AG-UI events | one trace per turn

Three routes at end of speech

- RETRIEVE: decompose, search, answer
- REFINE: delta search, claims updated in place
- SUPPRESS: restructure previous answer, 0 searches

Speculative answer

- Answer starts from the mid-utterance search while the decomposer runs
- Kept only if one reading reuses that search

Telemetry

- 27-field trace record per turn (JSONL)
- OpenTelemetry spans + metrics, 13-panel Grafana
- Cost includes priced local compute

Demo & Product Walkthrough

One command

```
docker compose up --build  
localhost:8000/?replay=1  
docker compose run --rm eval
```

Runs without an API key (deterministic offline arms);
recorded with gpt-4.1 + gpt-5.4-mini.

Typed input still streams

The console chunks text at 150 wpm, so the controller
runs identically for typed, replayed or spoken input.

Replay keys

1 compound 2 late detail 3 reformat 4
unanswerable R reset

1 Early retrieval + split

“Workshop in Pune for 30, cancellation
policy, catering”: search starts mid-sentence;
sub-query chips; 10 claims, 100% supported.

2 Ambiguity

“First quad jump?” splits into men's and
women's: Kurt Browning (1988), Miki Ando
(2002), both cited.

3 Refine, don't restart

“Big Ben's tower?” then “I meant since
2012”: Clock Tower rewritten to Elizabeth
Tower, no full-corpus search.

4 Suppression

“Repeat that in two bullets”: 0 searches,
original citations kept, first token in 1.46 s.

5 Grounding

Pets policy + parking price: nothing
verifiable, so the answer states uncertainty.
0 fabricated.

6 Telemetry

Timeline lead vector, v1→v2 diff, TTFT, cost
per step, support vs 85% target, Grafana.

Tools and tech stack used

Engine

- Python 3.11, plain asyncio turn loop
- FastAPI + Uvicorn: WS /stream, POST / agui (SSE)
- AG-UI protocol, JSON-Patch shared state

Retrieval

- rank-bm25 + NumPy in-process dense search
- BAAI/bge-small-en-v1.5 embeddings
- RRF + ms-marco-MiniLM-L-6 cross-encoder

LLMs & grounding

- gpt-4.1 decomposer, gpt-5.4-mini answers
- NLI deberta-v3-xsmall + MiniCheck-RoBERTa-L
- Circuit breaker, offline twin per model step

Operator console

- Next.js 16 static export, React 19, TypeScript
- Zustand, fast-json-patch, @ag-ui/core + zod
- Tailwind tokens, served by the engine

Evaluation

- ASQA: 5,301 samples, 15,679 chunks
- 6 gates, batch baseline, 4 ablations
- Independent NLI auditor; RAGAS harness

Ops & quality

- Docker Compose: one image, one command
- OpenTelemetry, Prometheus, Grafana
- 142 backend + 32 frontend test functions

Impact & Use case

0

fabricated citations shipped in 142 turns; 202 claims, each with a resolvable source

+20.2 pts

passage recall vs batch RAG on ASQA (85.5% vs 65.3%)

3.2 s

median head start on multi-clause requests;
88.9% / 87.6% of turns search before speech ends

Voice assistants & helpdesks

Search while the user is still describing the problem; corrections refine instead of restarting.

Contact-centre agent assist

Policy snippets surface as the caller speaks; every suggestion carries a checkable citation.

On-prem & privacy-sensitive

Embedder, reranker and verifiers run locally on CPU; only 2 LLM calls leave, to any OpenAI-compatible endpoint.

Regulated content

Unsupported claims become uncertainty, never fact; the per-turn trace is an audit trail.

Wasted work avoided: a suppress turn costs \$0.0003 and a refine turn \$0.0015, vs \$0.0059 for a full retrieve turn. Use cases are intended fits, not evaluated deployments.

Innovation highlights, results and limitations

Innovation highlights

- **Speculative retrieval** with a topic-shift guard that cancels stale searches (16 in the run)
- **Speculative answering** starts from the mid-utterance search while the decomposer runs
- **Coverage-guaranteed fusion:** every sub-query keeps its top 3 chunks
- **Validated streaming:** sentence released only after citation + NLI check
- **Claim-level refinement** with lineage: preserved and rewritten claims, no full search
- **Nearest-neighbour few-shot** decomposition: ASQA G3 22 → 30 of 40

Results: all 6 gates pass

Enterprise / ASQA (threshold)

G2 Early retrieval **88.9% / 87.6%** (≥ 80)

G3 Multi-intent **75% / 75%** (≥ 70)

G4 Citation support **93.3% / 86.9%** (≥ 85)

G5 Refinement **100% / 100%** (all)

G6 Telemetry **100% / 100%** (100)

Fabricated shipped **0 / 0** (0)

0% false triggers on 19 no-retrieve turns; audit model agrees with 98.2% / 93.2% of shipped claims.

Limitations

- **Slower first token:** median 3.6 s / 5.2 s vs 2.3 s / 2.2 s batch; p90 40 s from CPU verification
- **Cost 3-5× the baseline,** 67% of it the gpt-4.1 decomposer
- **Thin margins, one run:** G3 75 vs 70, ASQA G4 86.9 vs 85
- **Simulated speech:** no ASR; median lead on short questions only ~0.26 s
- **G5 checks the mechanism,** not answer quality; 4 of 18 refinements rewrote a claim
- **Small synthetic enterprise set** (22 turns)

What's next

Near term

- Bound verification latency: capped attribution search, GPU host; target p90 TTFT < 10 s
- Speculate less: 60 of 105 turns paid for a dropped third LLM call
- Real microphone with streaming ASR and partial-hypothesis revisions

Mid term

- Raise refinement yield: retry merged edits as single-block edits
- Multilingual cues and embedder, including Indian languages
- Multi-topic session memory, so older answers stay refinable

Longer term

- Several runs with confidence intervals; RAGAS scores; larger human-written set
- ANN index beyond $\sim 10^5$ chunks; concurrency and load testing
- Barge-in, streaming TTS, on-device small LLM

Differentiation

Claims a judge can check

1

Honest evaluation: 6 gates, same-parts baseline, 4 ablations, 92 failures analysed

2

Zero fabrication by construction + an independent auditor model

3

Rules as tests: no agent framework, no gold answer in src/, no inline prompts

4

Decision log D1-D35, including reversed and negative results

5

Standards-based wire: every frame is an AG-UI event; POST /agui serves any client

6

Observability: 27-field trace, OTel spans + metrics, 13-panel Grafana

Built to survive a real deployment

Circuit breaker

3 failures open it for 30 s;
offline twin per step

Injection fencing

evidence fenced as data, tags
neutralised

No API key needed

deterministic offline arms pass
the suite

Console = instrument

lead vector, v1→v2 diff, red
“unverified”

Accessible UI

no colour-only state, aria-live,
reduced motion

Corpus-agnostic

txt, md, json, jsonl, pdf;
content-addressed ids

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N) **Y**
- README with reproducible setup instructions (Y/N) **Y**
- Demo video, max 5 minutes (YouTube or Drive link)
- Presentation file (PPT or PDF) (Y/N) **Y**

Backend: github.com/VaibhavNaiyar/Rag-bd | Frontend: github.com/VaibhavNaiyar/Rag-fd

Thank you